# Open and Challenging Problems in Parallel Mechanisms

## Executive summary

Parallel mechanisms (a.k.a. parallel robots, parallel kinematic mechanisms/manipulators) remain scientifically rich because core tasks—workspace certification, singularity avoidance, calibration, and robust deployment—reduce to solving and reasoning about high-dimensional, nonlinear, often multi-solution constraint systems under uncertainty. A seminal "open problems" perspective from Jean-Pierre Merlet[1] already emphasized themes that are still central today: guaranteed/direct kinematics, singularities, design synthesis, obstacle avoidance, and calibration. [2]

Over the last decade, progress has been strong in specialized methods (e.g., certified kinematic reasoning under design uncertainties; convex/SOS relaxations for singularity-free workspace approximations; learning-augmented solvers and compensation), yet general, scalable, **guaranteed** solutions remain scarce. This is because the "full" problem typically couples (i) multiple assembly modes and changing topology of solution sets, (ii) mixed continuous/discrete design variables, (iii) geometric and non-geometric error sources (elasticity, thermal drift, backlash, friction), and (iv) application-specific constraints (collision avoidance, safety, metrology limitations). [3]

Ten open/challenging problems stand out as repeatedly encountered across planar, spatial, cable-driven, compliant, and hybrid parallel–serial architectures:

1. **Certified workspace and singularity-free region computation under uncertainty** (still limited to special classes, simplified constraints, or high compute). [4]
2. **Calibration that is observability-aware, low-cost, in-situ, and traceable** (hard because measurement is expensive, models are ill-conditioned, and non-geometric effects dominate in practice). [5]
3. **Real-time forward kinematics and assembly-mode tracking** for general mechanisms, especially with flexibility (sagging cables, compliant joints) and multiple feasible solutions. [6]
4. **Type synthesis/enumeration that is complete, scalable, and avoids duplicates**, especially for lower mobility / fewer-input-more-output and reconfigurable platforms. [7]
5. **Design optimization that jointly handles singularities, calibration, stiffness/dynamics, and manufacturability** (nonconvex, mixed-integer, topology-changing). [8]

6. **Learning-augmented synthesis/analysis with guarantees** (data scarcity + no formal safety/certification for learned surrogates). [9]
7. **Industrial machining adoption** (high stiffness/precision potential but limited uptake due to modeling complexity, small workspace, and calibration/compensation burden). [10]
8. **Medical imaging/surgery and rehabilitation** (MRI constraints, safety, human factors, and validation protocols remain difficult). [11]
9. **Cable-driven large-workspace parallel robots** (model uncertainty from sagging, elasticity, friction; tension feasibility; robust control). [12]
10. **Micro/nano and compliant parallel mechanisms** (high-bandwidth precision but dominated by compliance/hysteresis/fatigue and difficult calibration). [13]

# Framing and taxonomy

## What makes parallel-mechanism problems hard

Most analysis/design questions can be written as a constrained system

$$F(x, q, p) = 0$$

where $x$ is end-effector pose (often SE(3) or a lower-mobility subset), $q$ is actuator/joint space, and $p$ are design and error parameters. For parallel mechanisms, forward kinematics is frequently multi-solution and can change solution count across the workspace (assembly modes, working modes), while singularities form complicated sets that can separate regions of qualitatively different behavior. [14]

A second complication is that *application-grade* questions add constraints beyond ideal kinematics: joint limits, collisions, manufacturing tolerances, elastic deflection, thermal/gravity-induced errors, sensor/actuator offsets, and (for cables) tension positivity and sagging models. These transform "solve $F = 0$" into a hybrid of nonlinear algebraic geometry, global optimization, and uncertainty quantification. [15]

## Taxonomy of open problems and dependencies

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flowchart TD
  A[Parallel mechanism problem] --> B[Theoretical analysis]
  A --> C[Design methodology]
  A --> D[Practical deployment]

  B --> B1[Workspace & singularity certification]
  B --> B2[Calibration / identification]
  B --> B3[Forward kinematics & mode tracking]
  B --> B4[Stiffness/dynamics w/ uncertainty]
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C --> C1[Type synthesis / enumeration]
C --> C2[Multiobjective optimization]
C --> C3[Learning-augmented synthesis/analysis]

D --> D1[Industrial machining & accuracy]
D --> D2[Medical/rehab safety & validation]
D --> D3[Cable-driven large workspace]
D --> D4[Micro/nano compliant PMs]

B1 --> C2
B2 --> D1
B3 --> D3
B4 --> D4
C2 --> D1
C3 --> B3
```

This dependency structure is repeatedly observed in the literature: e.g., certified workspace reasoning used to constrain design/control; calibration and error mapping required for machining; forward-kinematics solvers and mode tracking central for cable-driven systems; and stiffness/compliance models essential for micro/precision mechanisms. [16]

## Open problems in theoretical analysis

### Certified workspace and singularity-free region computation under uncertainty

**Problem definition (precise).** Given a mechanism architecture and parameters $p$ (nominal + tolerances), compute (i) a reachable workspace set $W \subseteq \mathcal{X}$, (ii) subsets with guaranteed properties such as "singularity-free" and "unique assembly mode," and (iii) bounds on performance measures (conditioning, transmissibility, stiffness) over $W$, **with certificates** accounting for parameter uncertainty and numerical instability. [17]

**Why it remains unsolved.** Even without uncertainty, exact workspace/singularity sets can be high-dimensional semi-algebraic sets with intricate topology; adding tolerances turns this into robust set computation ("for all $p \in \mathcal{P}$") that is computationally prohibitive in general. A contemporary dedicated effort explicitly frames *certification* (preventing both computational and physical singularities, with design uncertainties) as a major methodological objective, underscoring that non-certified sampling is insufficient for safety-critical control guarantees. [18]

**Technical challenges.** The core difficulties are (a) mixed trigonometric/polynomial constraints and high DOF pose spaces; (b) nonconvexity, disconnected feasible sets, and multiple solution branches; (c) robust handling of tolerances/intervals and floating-point issues; (d) incorporation of joint limits, collision constraints, and

(for cables) unilateral tension constraints; and (e) computational scaling with mechanism complexity. [19]

**Current approaches and limitations.**
Certified/symbolic–numeric approaches combine symbolic computation (e.g., discriminant varieties, parametric system reasoning) with continuation and interval calculus to "certify" kinematic behavior under uncertainty, but they are often limited by computational cost and by assumptions such as generically zero-dimensional solution sets and rigid, nonredundant mechanisms. [20]
Convex relaxations can approximate singularity-free workspaces using sum-of-squares (SOS) programming and semidefinite programming, producing inner/outer ellipsoid approximations, but these require polynomialization steps and typically yield conservative approximations that may not capture nonconvex geometry well. [21]
Interval-analysis methods can compute maximal inscribed singularity-free circles/spheres (for planar/spatial examples), giving systematic procedures, but they do not by themselves solve the full 6D "task-relevant" workspace certification problem for general mechanisms. [22]

**Key mathematical models / assumptions.** Typical assumptions include rigid-body kinematics, exact loop-closure constraints, and singularities detected via Jacobian rank conditions (Type-I/Type-II or equivalent classifications), while uncertainty is modeled by intervals on $p$ or bounded perturbations. For SOS methods, kinematics must be expressed as polynomial constraints (often by introducing additional variables and constraints to replace trigonometric functions). [23]

**Promising research directions.** A practical path is hybridization: certified symbolic–numeric pipelines to localize "hard" boundary/singularity regions + convex/SOS or interval methods to produce conservative certified safe sets for control, together with explicit uncertainty budgets. Another underdeveloped direction is co-design of architecture + certification: choose mechanism forms whose kinematics admit tractable certification (low-degree polynomials, sparsity, symmetry). [24]

**Evaluation metrics.** Candidate metrics include (i) certified volume (or task-measure) of safe workspace; (ii) tightness gap between certified inner/outer approximations; (iii) worst-case distance-to-singularity bounds; (iv) computation time vs. mechanism size; (v) robustness margins under specified tolerance sets; and (vi) false-safe and false-unsafe rates (when compared to high-fidelity simulation or measurement). [23]

**Benchmarks / datasets.** Widely used benchmark mechanisms include Gough–Stewart/hexapods, Delta-type translational robots, spherical parallel manipulators, and representative lower-mobility mechanisms appearing in workspace/singularity studies. A gap is the lack of standardized, open benchmark suites with ground-truth certified sets across architectures. [25]

## Calibration that is observability-aware, low-cost, in-situ, and traceable

**Problem definition (precise).** Estimate and compensate parameter errors $\delta p$ (geometric and increasingly elastogeometric/non-geometric effects) so that pose error $\| x_{\text{pred}}(q, p + \delta p) - x_{\text{meas}} \|$ is minimized (or bounded) over a task workspace, under realistic measurement noise, limited sensing, and potentially changing conditions (load, temperature). Modern variants explicitly target robustness to ill-conditioning and unit/scale issues in identification matrices. [26]

**Why it remains unsolved.** In practice, calibration is limited by (i) expensive metrology (laser trackers, photogrammetry, optical systems), (ii) identifiability/observability issues—many parameters are weakly observable or correlated in parallel architectures—and (iii) dominance of non-geometric effects (compliance, thermal drift, joint clearances) that violate rigid-model assumptions. Even improved formulations that reduce ill-conditioning still rely on careful measurement design and may not generalize across the whole workspace. [27]

**Technical challenges.** Core challenges are (a) designing excitation/measurement poses that maximize observability without entering singularities; (b) coping with multi-closed-loop coupling and parameter correlation; (c) integrating compliance/loads into calibration in a stable way; (d) achieving repeatable, low-cost metrology; and (e) providing uncertainty quantification/traceability so the compensated accuracy is credible for manufacturing or medicine. [28]

**Current approaches and limitations.**
Model-based calibration remains strong: for example, a 5-axis parallel machining robot calibration method uses dual quaternions for forward kinematics plus a dimensionless error mapping approach to mitigate unit-related ill-conditioning, validated with full-pose measurements via laser tracker and showing large residual-error reductions on a prototype. [29]
Model-free or “modeless” compensation approaches exist (e.g., split calibration concepts), but they risk weaker extrapolation, poorer physical interpretability, and difficulty handling configuration changes. [30]
Digital-twin and learning-based compensation frameworks are emerging for parallel robots, but they depend on data collection and still face metrological trust/uncertainty and distribution-shift issues. [31]
Patents also reflect industrial motivation for simplified calibration procedures (e.g., calibrating delta robots without extensive disassembly; closed-chain calibration concepts “without external measurement equipment”), but patent disclosures often stop short of open, benchmarked performance evidence. [32]

**Key mathematical models / assumptions.** Common modeling choices include differential error models (Jacobian-based linearization), dual-quaternion or SE(3) representations to avoid rotation singularities, least squares or robust norms (e.g., $L_\infty$) for parameter estimation, and separation of error sources into geometric vs.

non-geometric components. Identification often assumes quasi-static measurement conditions and sufficiently accurate pose ground truth. [33]

**Promising research directions.**
A high-impact direction is "calibration as an experiment design + inference problem" with explicit observability criteria, robust estimators, and uncertainty propagation to certified accuracy envelopes. A second is hierarchical calibration: (i) fast factory geometric calibration, (ii) in-situ self-check via embedded sensors and constraints, (iii) online adaptation for drift (thermal/load). A third is standardized reporting: publish not only before/after errors but also uncertainty budgets and generalization tests across withheld workspace regions. [34]

**Evaluation metrics.** Useful metrics include RMS and worst-case pose error over a test set; generalization error to unseen workspace regions; parameter uncertainty (covariance/credible intervals); stability across loads/temperature; calibration time and required instrumentation; and reproducibility across repeated calibrations. [35]

**Benchmarks / datasets.** A notable gap is publicly shared datasets for *parallel* robot calibration with high-accuracy ground truth. Related robotics calibration datasets (e.g., robot–world/hand–eye calibration datasets on public repositories; camera–robot calibration datasets such as METRIC) exist and can inspire protocol design but do not fully represent parallel-robot loop-closure specifics. [36]

## Real-time forward kinematics and assembly-mode tracking for general and flexible parallel mechanisms

**Problem definition (precise).** Given actuator values $q$, compute all feasible poses $x$ (or a physically consistent subset) satisfying loop closure, and track the correct branch over time under perturbations, while avoiding singularities and numerical instability. For flexible/cable-driven systems, the "forward problem" includes additional variables (cable tensions, sag parameters) and feasibility constraints (positive tension, stability). [37]

**Why it remains unsolved.** Direct/forward kinematics can be computationally heavy and multi-solution; selecting the right solution branch robustly in real time is still difficult, especially near singularities or when the model includes non-algebraic components (e.g., catenary sag). Foundational work already highlighted "guaranteed" direct-kinematics solving as an open problem; modern cable-driven and learning-augmented methods reinforce that general-case solutions remain challenging. [38]

**Technical challenges.** These include (a) explosion of solution count and branch switching; (b) sensitivity near singularities and poor conditioning; (c) modeling fidelity vs. speed tradeoffs; (d) ensuring physical feasibility (stability, tension positivity) in cable-driven robots; and (e) real-time integration with estimation/control under noise. [39]

**Current approaches and limitations.**
Numerical continuation, including monodromy-based strategies, has been demonstrated for forward kinematics of cable-driven robots with sagging cables and benchmarked against existing approaches, but such methods can still be complex to implement robustly and may not provide formal real-time guarantees in all regimes. [40]
Hybrid methods mixing neural networks with continuation/symbolic computation aim to accelerate solving parametric nonlinear systems, explicitly motivated by multi-solution structure; however, correctness guarantees and failure modes remain key limitations, especially for safety-critical control. [41]
For cable-driven robots, earlier work emphasizes the general-case difficulty (multiple solutions, stability, positive tensions) under sagging models, suggesting substantial remaining headroom. [42]

**Key mathematical models / assumptions.** Standard rigid parallel robots rely on algebraic loop-closure equations; cable-driven models may use catenary-based relations (nonlinear, sometimes non-algebraic) plus static equilibrium. Many solvers rely on local convergence assumptions (good initializations) and smoothness away from singularities. [39]

**Promising research directions.**
A credible direction is *certified real-time pipelines*: (i) offline compute branch structure and safe regions (certified), (ii) online track within certified regions using fast local solvers, (iii) detect approaching singularities with provable margins and re-plan. Another is solver benchmarking standards: same mechanism, same tolerances/noise, compare speed, convergence basin, and failure probability. [24]

**Evaluation metrics.** Trackable metrics include solve time distributions, convergence basin size, probability of wrong-branch selection under noise, safety margin to singularities, and closed-loop tracking error in dynamic experiments. [43]

**Benchmarks / datasets.** Promising benchmarks include cable-driven setups with sagging and limited tension; and representative high-DOF platforms where multi-solution behavior is common. Public datasets for “FK with ground truth branch labels” are rare and would substantially strengthen reproducibility. [44]

## Open problems in design methodology

### Complete and scalable type synthesis and enumeration across modern mechanism families

**Problem definition (precise).** Given a desired motion pattern (e.g., 3T, 2T1R, mixed DOF, fewer-input–more-output), systematically enumerate non-isomorphic parallel architectures (including hybrid parallel–serial and reconfigurable forms) whose constraint/actuation screws achieve the task mobility, while controlling overconstraint, singularities, and practical realizability. [45]

**Why it remains unsolved.** The combinatorial state space of architectures grows quickly; equivalence (isomorphism) checking is difficult; and practical filters (manufacturability, calibration, safety, collision) require quantitative evaluation beyond pure mobility. Recent work continues to propose new synthesis concepts (e.g., "separation of constraints and drives" to reduce drive count), indicating that no single unified methodology dominates across families. [46]

**Technical challenges.** Key obstacles are (a) formal representation of constraints/actuation (screw theory, Lie groups, constraint graphs) across mixed DOF; (b) completeness without duplicates; (c) incorporation of overconstraint and redundancy; (d) bridging abstract synthesis and detailed dimensional design; and (e) lack of shared "structure databases" openly available for general use. [47]

**Current approaches and limitations.** Screw-theory-based type synthesis remains prevalent and can generate families across motion types, but completeness and scalability depend heavily on problem-specific representations and human-guided filtering. [48]
Hybrid-robot reviews show strong growth in parallel–serial architectures and highlight that many synthesis/analysis papers still focus mainly on the parallel submechanism, neglecting full-system co-design. [49]
Machine-learning approaches to mechanism synthesis are emerging, but they typically require datasets and can struggle with ensuring physical correctness and handling discrete topology choices. [50]

**Key mathematical models / assumptions.** Many synthesis methods assume rigid joints and ideal constraints and use screw-system rank conditions or group-theoretic representations to match desired DOF. Practical designs then must address singularities and conditioning, which are not captured by mobility alone. [51]

**Promising research directions.** A promising direction is "computational synthesis with certificates": generate candidate architectures via constraint-based enumeration, then prune using certified workspace/singularity checks and calibratability criteria. Another is standardized representations (graph + screw descriptors) to enable open, shareable structure databases and reproducible enumeration benchmarks. [52]

**Evaluation metrics.** Metrics include coverage (fraction of known families recovered), uniqueness (duplicate rate), compute scaling with link/joint count, and downstream performance distributions (workspace volume, stiffness, conditioning) of generated architectures. [53]

**Benchmarks / datasets.** Proposed benchmark artifacts: open canonical structure datasets for lower-mobility and hybrid robots; and "synthesis challenges" specifying motion patterns plus practical constraints (drive count, joint types,

obstacle constraints). Current open datasets are limited, especially for parallel architectures. [54]

## Singularity- and calibratability-aware multiobjective design optimization with topology changes

**Problem definition (precise).** Optimize geometric parameters (and sometimes topology) to satisfy task constraints while maximizing performance indices (dexterity, stiffness, accuracy, dynamic response) and explicitly avoiding singularities and uncalibratable regions. This is typically a mixed continuous/discrete, nonconvex, multiobjective problem under constraints, often with discontinuities when configuration-space topology changes. [55]

**Why it remains unsolved.** The objective landscape can have flat regions and abrupt jumps driven by topology changes (e.g., new singularities appearing/disappearing), making gradient-based methods unreliable. A concrete modern example shows that singularity-aware design optimization must actively detect singularities in the targeted operation range and that topology changes complicate derivative computation, motivating stochastic optimization. [56]

**Technical challenges.** The hard parts are (a) evaluating singularities globally over a task region; (b) incorporating calibration observability and stiffness/dynamics models without making the optimization intractably slow; (c) managing discrete architecture choices and continuous parameters together; and (d) ensuring that optimized designs are robust to manufacturing tolerances and modeling errors. [57]

**Current approaches and limitations.**
Singularity-aware optimization frameworks exist that combine inverse-kinematics sampling with adversarial detection of forward/backward singularities, but they often rely on stochastic methods (e.g., PSO variants) with no global optimality guarantee and substantial compute. [58]
Performance-index-driven optimization experiences well-known issues: e.g., classical indices like condition numbers can be misleading for mixed translation–rotation DOF due to unit inhomogeneity, and even defining global indices can be computationally intensive and not yet "certified." [59]
Industrial reviews of parallel machine tools stress that complex modeling and limited workspace remain barriers to uptake, implying that current design pipelines do not yet deliver robust turnkey solutions. [60]

**Key mathematical models / assumptions.** Design optimization typically assumes rigid-body kinematics plus Jacobian-based singularity measures and chosen performance indices (manipulability, transmissibility, stiffness). When stiffness is included, virtual-joint methods and nonlinear stiffness aggregation models under internal/external loadings are common, but incorporating these inside optimization increases complexity. [61]

**Promising research directions.**
A strong direction is "calibratability as a design objective," integrating observability measures into optimization (pose selection + parameter sensitivity) alongside singularity avoidance. Another is hierarchical optimization: coarse topology search (enumeration/pruning) → mid-level certified workspace screening → fine dimensional optimization with high-fidelity stiffness/thermal models. [62]

**Evaluation metrics.** Use Pareto-front quality (hypervolume), robustness (performance under perturbed parameters), certified singularity-free task coverage, calibration generalization error, and computational cost. [63]

**Benchmarks / datasets.** Benchmarks should include multiobjective tasks (e.g., prescribed machining trajectories or rehab trajectories) with explicit forbidden singularity regions and standardized reporting of indices and robustness. Emerging optimization-for-safety examples in mechanism design suggest feasible templates but require robotics-specific benchmark suites. [64]

## Learning-augmented analysis and synthesis with verifiable guarantees

**Problem definition (precise).** Use machine learning to (i) approximate kinematic maps (FK/IK), (ii) perform error compensation/calibration, and/or (iii) synthesize mechanisms/topologies, while ensuring physical feasibility, robustness, and (ideally) certified safety margins (e.g., no singularity violations). [65]

**Why it remains unsolved.** ML can produce fast approximations but lacks intrinsic guarantees; parallel mechanisms can have discontinuities (branch switching), extrapolation failures near singularities, and strong dependency on geometry/topology. Surveys on ML for mechanism kinematic analysis/synthesis and on ML in parallel-robot applications document rapid growth but do not resolve the guarantee problem. [66]

**Technical challenges.** The main barriers are (a) representing mechanism topology and constraints in a learnable yet physically faithful way; (b) data scarcity for diverse architectures; (c) handling multi-solution FK and branch identification; (d) quantifying uncertainty and detecting out-of-distribution inputs; and (e) integrating learned components into certified control pipelines. [67]

**Current approaches and limitations.**
"Neural-network-as-synthesis-engine" ideas map mechanism topology to a trainable network, reframing synthesis as gradient-based learning and demonstrating applicability across planar/spatial/origami generalized parallel mechanisms, but the method's completeness, correctness guarantees, and dependence on training/initialization remain open issues. [68]
Learning can also be combined with deterministic solvers (e.g., neural + continuation/symbolic computation) for nonlinear parametric systems, explicitly motivated by the need to manage multiple solutions, but formal worst-case

guarantees are still not standard. [41]
In calibration, data-driven digital-twin compensation for parallel robots is being explored, yet metrological trust and generalization remain central concerns. [31]

**Key mathematical models / assumptions.** ML approaches often assume that training data covers relevant configurations and that a single-valued regression is meaningful; but for parallel mechanisms FK is frequently set-valued. Therefore, mixture models, implicit neural representations, or classification+regression (branch ID + local solution) are often needed. [69]

**Promising research directions.** The most promising path is **certified learning**: (i) use ML to propose candidate solutions/initializations or to learn residual error fields, (ii) verify/correct with certified numerical methods (interval/continuation/SOS constraints), and (iii) maintain runtime monitors that detect approaching singularities and distribution shifts. [70]

**Evaluation metrics.** Beyond accuracy/latency, evaluations should measure failure probability near singularities, conformity to constraints, uncertainty calibration quality, and impact on closed-loop safety and task success. [71]

**Benchmarks / datasets.** A key missing element is open datasets that include (a) geometry/topology descriptors, (b) multi-solution FK labels, and (c) ground-truth safety annotations (distance-to-singularity, collision-free labels). Existing generic robot calibration datasets can be repurposed for method development but do not close the parallel-robot gap. [72]

## Open problems in practical applications

### Industrial machining and machine tools: accuracy under load/thermal conditions with manageable calibration burden

**Problem definition (precise).** For parallel kinematic machine tools and machining robots, achieve specified volumetric accuracy (and stiffness) over a task workspace under machining loads and thermal/gravity effects, with a calibration/compensation pipeline that is economically viable in industrial cycles and robust over time. [73]

**Why it remains unsolved.** Even though closed-loop kinematics can provide stiffness and precision advantages, industrial uptake is still limited relative to serial machines, with identified barriers including complex modeling and small workspace. Calibration and error compensation are emphasized as critical yet demanding. [60]

**Technical challenges.** The main issues are (a) elasto-geometric deflection under cutting loads; (b) thermal drift and gravity-induced deformation; (c) maintaining accuracy across the workspace despite varying stiffness; (d) integrating calibration

into production without costly metrology downtime; and (e) guaranteeing singularity-free, well-conditioned toolpaths. [73]

**Current approaches and limitations.** Advanced kinematic calibration methods for machining-oriented parallel robots demonstrate large improvements in residual pose errors via improved modeling and measurement, but they rely on specialized metrology (e.g., laser trackers) and careful experimental procedures. [29] Workspaces and performance indices are widely used in design and evaluation, but as emphasized in performance-measure reviews, global indices can be costly and local indices may be misleading near singularities or for mixed DOF without careful normalization. [59]

**Key mathematical models / assumptions.** Industrial pipelines often assume a calibrated geometric model plus add-on compensation fields; stiffness is modeled via virtual-joint methods and/or nonlinear stiffness aggregation under internal/external loadings; dynamic effects may be included for high-speed machining. [74]

**Promising research directions.** Integrate metrology-grade digital twins with uncertainty propagation (so compensation is traceable), and co-design calibration observability with mechanism design (choose architectures and sensor placements that are inherently calibratable). Also prioritize open benchmark tasks (e.g., standard machining trajectories) for reproducible evaluation of calibration + compensation pipelines. [75]

**Evaluation metrics.** Volumetric error over standardized test trajectories, stiffness/deflection under load, drift over time/temperature, calibration time/cost, and rate of re-calibration required. [76]

**Benchmarks / datasets.** Benchmarking is often done on representative hexapods/hybrid machining architectures; however, open datasets that include pose measurements under load and thermal conditions remain sparse. [60]

## Medical imaging/surgery and rehabilitation: safety validation, compatibility constraints, and clinical-grade evidence

**Problem definition (precise).** Design and deploy parallel mechanisms that provide precise motion in constrained clinical environments (e.g., MRI), or that safely interact with humans for rehabilitation/assistance, while meeting stringent safety, compatibility, and validation requirements. [77]

**Why it remains unsolved.** Clinical contexts impose constraints that are not present in industrial settings: MRI compatibility (materials, actuation, sensing), sterility, limited space, patient safety, and high evidentiary burden. A dedicated 6-DoF parallel robot for MR environments illustrates the need for careful design, workspace analysis, and experimental validation in scanner conditions. [78]

For rehabilitation and humanoid/assistive applications, a systematic review covering 2012–2023 highlights broad activity but also underscores the need for clear performance evaluation methods and the presence of persistent technological challenges. [79]

**Technical challenges.** Key challenges are (a) guaranteeing singularity-free operation for patient-safe trajectories; (b) compliance and backdrivability for safe contact; (c) accurate sensing/estimation in constrained environments (MRI); (d) calibration and repeatability verification with clinical-grade protocols; and (e) translating prototypes to higher technology readiness levels (TRLs). [80]

**Current approaches and limitations.** Reviews of computer-aided fracture reduction robotics describe progress toward clinical use but highlight unresolved gaps, ethics, learning curves, and system-level challenges—parallel mechanisms appear prominently due to their stiffness and controllability, yet validation remains substantial work. [81]
MRI-focused parallel robot designs are being experimentally evaluated, but dissemination of standardized open datasets and consensus performance protocols is limited. [78]

**Key mathematical models / assumptions.** Safety typically relies on workspace/singularity analysis, force/torque limits, and control constraints; compliance can be modeled with compliance matrices or virtual-joint methods. In MRI settings, actuation/sensing constraints frequently require nonstandard mechatronics, increasing model uncertainty. [82]

**Promising research directions.** Develop shared validation protocols (accuracy, repeatability, safety margins) and publish open benchmark datasets for representative clinical tasks (needle placement, fracture reduction trajectories, rehab exercises). Combine certified workspace reasoning with runtime monitoring to enforce safety envelopes. [83]

**Evaluation metrics.** Clinical-task error (targeting error), repeatability, safety incidents/near-misses, force-limit compliance, procedure time, and robustness across patients/anatomical variability. [84]

**Benchmarks / datasets.** Current benchmarking is fragmented; establishing shared datasets analogous to general robotics calibration datasets would be transformative for reproducible progress in parallel clinical robots. [85]

## Cable-driven large-workspace parallel robots: modeling uncertainty, feasibility, and robust control

**Problem definition (precise).** For cable-driven parallel robots (CDPRs), compute and control motion while enforcing tension feasibility (typically positive tensions

and sometimes bounded by hardware), accounting for sagging/elasticity and friction, and maintaining accuracy across large workspaces. [86]

**Why it remains unsolved.** Cables introduce strong model uncertainties (sagging, elasticity, nonlinear friction), making it difficult to eliminate uncertainty from control design; reviews emphasize CDPR potential but also their modeling difficulty compared to rigid-limb parallel robots. [87]
Flexible-cable forward kinematics in the general case is explicitly recognized as difficult; more recent work proposes advanced continuation-based approaches to solve FK faster, indicating ongoing challenge. [88]

**Technical challenges.** The main difficulties are (a) non-algebraic/nonlinear cable models; (b) multiple FK solutions and stability constraints; (c) sensitivity to parameter drift (cable creep, temperature); (d) coupling between tension distribution and pose; and (e) robust control under uncertainty without sacrificing performance. [89]

**Current approaches and limitations.** Control-strategy reviews catalog numerous robust/adaptive methods and note persistent uncertainties; while effective in experiments, methods can depend on tuning and may not generalize across configurations/payloads. [87]
Kinematic calibration methods exist for CDPRs (including pulley kinematics), but calibration itself inherits the same uncertainty and measurement constraints. [90]
Learning-based FK modeling for CDPRs is emerging, but benchmarked generalization and safety guarantees remain open. [91]

**Key mathematical models / assumptions.** Models range from simplified massless straight cables to catenary sag models with static equilibrium; feasibility often includes unilateral tension constraints. Many control strategies assume bounded uncertainty and rely on adaptive/robust control frameworks. [89]

**Promising research directions.** Combine certified kinematic tools with feasibility analysis (tension bounds) to produce guaranteed safe operational envelopes; develop benchmark CDPR datasets with ground-truth pose and tension measurements; and evaluate solver/controller robustness under controlled perturbations (payload changes, cable length errors). [92]

**Evaluation metrics.** Feasible workspace volume under tension constraints, tracking error under payload variation, FK solver failure rate, and robustness margins to parameter drift (e.g., cable elasticity changes). [93]

**Benchmarks / datasets.** CDPR benchmarks exist in research labs, but open datasets for sagging-cable FK/IK with physical feasibility labels remain rare. [42]

## Micro/nano and compliant parallel mechanisms: precision limited by compliance, hysteresis, and fatigue

**Problem definition (precise).** Achieve high precision and bandwidth in micro/nano positioning via flexure-based or compliant parallel mechanisms, while modeling and compensating compliance, parasitic motion, and hysteresis, and ensuring long-term reliability under cyclic loading. [94]

**Why it remains unsolved.** In compliant mechanisms, the "mechanism" is often the elastic structure itself; kinematics is inseparable from mechanics, so rigid-body assumptions break down. Compliance modeling is feasible but becomes complex for full 6-DOF flexure-based Stewart-platform-like micromanipulators, and calibration at sub-micron levels is experimentally challenging. [95]

**Technical challenges.** Key issues include (a) accurate compliance/stiffness modeling with parasitic motion; (b) parameter drift and hysteresis; (c) high-sensitivity metrology at micro scales; (d) fatigue and manufacturing variability; and (e) integrating these models into real-time control. [96]

**Current approaches and limitations.** Compliance matrix methods and related modeling techniques provide systematic models, but parameter identification and validation remain costly. [97]
Broader stiffness modeling work (virtual-joint methods, non-linear stiffness under load) informs design, but translating these to micro-scale and to manufacturing variability is still difficult. [98]
Emerging micro-robot fabrication (including miniaturized classical parallel mechanisms via novel fabrication paradigms) expands the design space, creating additional need for standardized modeling and calibration practices. [99]

**Key mathematical models / assumptions.** Strain-energy-based stiffness modeling, compliance matrices, and virtual-spring lumped models are common; many approaches still assume quasi-static conditions and linear material behavior in restricted ranges. [100]

**Promising research directions.** Co-design geometry and sensing (embed strain/deflection sensing), develop standardized calibration artifacts (micro-scale reference trajectories and fixtures), and report uncertainty budgets. Also, use multi-fidelity modeling: FE-based design-time validation + reduced-order real-time models for control. [101]

**Evaluation metrics.** Nanometer/micron-level positioning accuracy, hysteresis compensation performance, drift over time/cycles, fatigue life, and repeatability under variable loading. [102]

**Benchmarks / datasets.** Public datasets are scarce; a practical benchmark set could include compliance-identified parameters plus repeated trajectory data under controlled thermal and loading conditions. [102]

## Cross-cutting comparison, benchmarks, and research agenda

### Comparative table of open problems across theory, design, and applications

| Problem name | Description | Challenges | Current methods | Gaps | Suggested directions |
|---|---|---|---|---|---|
| Certified workspace & singularity-free regions | Compute guaranteed safe workspace subsets under tolerances/uncertainty | High-dimensional semi-algebraic sets; multi-branches; robustness; numerical stability | Certified symbolic–numeric + intervals; SOS inner/outer approximations; interval maximal spheres | Poor scaling; conservatism; limited collision/joint-limit integration | Hybrid certification + convex/interval; architecture co-design for certifiability; open benchmark suites [4] |
| Observability-aware calibration (in-situ, traceable) | Estimate geometric + non-geometric errors with credible uncertainty bounds | Ill-conditioning; correlated parameters; expensive metrology; drift under load/thermal | Dual-quaternion/SE(3) error models; robust estimation; digital-twin compensation | Limited open datasets; weak uncertainty reporting; non-geometric dominance | Calibration experiment design + uncertainty propagation; hierarchical/online calibration; standardized reporting [103] |
| Real-time FK & | Solve FK and track correct | Multi-solution; | Continuation/monodromy | Formal guarantee | Certified offline |

| Problem name | Description | Challenges | Current methods | Gaps | Suggested directions |
|---|---|---|---|---|---|
| mode tracking | branch robustly (incl. flexible/cable models) | near-singularity instability; non-algebraic models for cables | FK; learning-augmented solvers; hybrid NN+deterministic | s rare; limited datasets with branch labels | partitioning + online tracking; failure monitoring; benchmarked solver comparisons [104] |
| Scalable type synthesis & enumeration | Enumerate architectures for target mobility (hybrid, lower-mobility, fewer-input-more-output) | Combinatorics; isomorphism; bridging to dimensional design | Screw-theory synthesis; hybrid-robot architecture surveys; emerging ML synthesis | Lack of open structure databases; incomplete/biased enumeration | Standard graph+screw representations; synthesis + certified pruning; open structure repositories [105] |
| Design optimization with singularities + calibratability | Multiobjective optimize geometry/topology with guaranteed task coverage | Nonconvexity; topology changes; expensive evaluation | Singularity-aware sampling + stochastic optimization; performance-index optimization | No global optimality; index pathologies; slow high-fidelity loops | Hierarchical multi-fidelity optimization; include calibratability & uncertainty; certify task coverage [8] |
| Learning-augmented synthesis/analysis | Use ML to accelerate FK/IK, compensation, synthesis | Data scarcity; multi-solution mapping; | Reviews and early methods (TMNN synthesis; | Safety/certification unsolved; standard datasets | Certified learning loops (ML propose + certified |

| Problem name | Description | Challenges | Current methods | Gaps | Suggested directions |
|---|---|---|---|---|---|
| with guarantees | | OOD; explainability | digital-twin compensation) | lacking | verify); uncertainty-aware models; benchmark datasets [65] |
| Industrial machining adoption | Deliver stiffness/precision benefits at scale with manageable modeling/calibration | Load/thermal effects; workspace limits; integration cost | Open-access industrial review; advanced calibration methods | Limited “turnkey” pipelines; sparse open under-load datasets | Metrology-traceable digital twins; standardized machining benchmarks; robust compensation [10] |
| Medical imaging & rehab deployment | Safe, validated parallel robots under MRI/clinical constraints | Compatibility; patient safety; validation; TRL progression | MRI 6-DoF platform experiments; rehab systematic review | Standard protocols/datasets limited | Publish task datasets + validation protocols; certified safety envelopes; human-factor co-design [11] |
| CDPR large-workspace robustness | Control with sagging/elasticity + unilateral tensions | Model uncertainty; feasibility; drift | Control-strategy review; FK methods for sagging cables; | Missing open feasibility-labeled datasets | Certified feasible workspace computation; |

| Problem name | Description | Challenges | Current methods | Gaps | Suggested directions |
|---|---|---|---|---|---|
| | | | calibration methods | | robust control benchmarking; shared datasets [12] |
| Micro/nano compliant PMs | Precision limited by compliance/hysteresis/fatigue | Model/identification; metrology; reliability | Compliance matrix models; stiffness modeling frameworks; emerging miniaturization | Calibration at scale; fatigue datasets | Embedded sensing co-design; standardized micro calibration artifacts; multi-fidelity control models [106] |

## Research attention over time: calibration as an example proxy

A coarse proxy for research attention is the number of scholarly works matching a topical query over time. Using OpenAlex[107] and grouping works matching the full-text search query **“parallel robot calibration”** by publication year for 2014–2025 yields the following counts (note: this is a keyword proxy, not a curated field-specific census, and later years may be incomplete due to indexing and publication delays). [108]

```
xychart-beta
  title "Works matching 'parallel robot calibration' by year (OpenAlex query proxy)"
  x-axis [2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025]
  y-axis "works" 0 --> 4200
  line [1058, 1146, 1406, 1494, 1861, 2184, 2643, 2969, 3107, 3803, 1921, 1034]
```

The general rise through 2023 is consistent with broader observations that performance indices, calibration, and compensation have become increasingly

central as parallel robots move into demanding applications (machining, high-speed pick-and-place, medical), even though top-level industrial adoption barriers persist. [109]

## Benchmark systems and dataset schema proposal

A major cross-cutting gap is the absence of standardized, widely adopted benchmarks that combine: geometry, kinematic constraints, uncertainty/tolerances, ground-truth measurements, and safety annotations (singularity proximity, collision-free labels). This gap is repeatedly felt in certified workspace computation, calibration reproducibility, and ML evaluation. [52]

A minimal dataset/entity schema that would support reproducible research is:

```
erDiagram
  MECHANISM ||--o{ GEOMETRY_PARAM : has
  MECHANISM ||--o{ TRAJECTORY : evaluated_on
  MECHANISM ||--o{ MEASUREMENT_SET : measured_by
  TRAJECTORY ||--o{ POSE_SAMPLE : contains
  POSE_SAMPLE ||--o{ SAFETY_LABEL : annotated_with
  MEASUREMENT_SET ||--o{ SENSOR_MODEL : uses
  MEASUREMENT_SET ||--o{ UNCERTAINTY_BUDGET : reports
  GEOMETRY_PARAM ||--o{ TOLERANCE : bounded_by

  MECHANISM {
    string architecture_id
    string topology_descriptor
    string dof_type
  }
  GEOMETRY_PARAM {
    string name
    float nominal_value
    string unit
  }
  TOLERANCE {
    float lower
    float upper
    string distribution_assumption
  }
  TRAJECTORY {
    string task_type
    string workspace_region
  }
  POSE_SAMPLE {
    float[] q
    float[] x_measured
    float[] x_predicted
  }
  SAFETY_LABEL {
```

```
    float dist_to_singularity
    bool collision_free
    string mode_id
  }
  MEASUREMENT_SET {
    string instrumentation
    string calibration_procedure
  }
  SENSOR_MODEL {
    string model
    float noise_params
  }
  UNCERTAINTY_BUDGET {
    float pose_uncertainty
    float param_uncertainty
  }
```

This schema is aligned with what modern calibration and compensation work increasingly needs—explicit measurement procedures, uncertainty propagation, and labels for safety/branch behavior—yet such datasets are rarely open for parallel mechanisms today. [110]

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